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(Deemed to be University under section 3 of UGC Act, 1956)

School of Advanced Sciences

Fall Semester 2024-2025

Continuous Assessment Test – I

Answer Key

Programme Name & Branch : B.Tech. Computer Science with all Specializations

Slot: **C2+TC2+TCC2**

Course Name & code: **Discrete Mathematics and Graph Theory & BMAT205L**

Class Number (s): **2538, 2559, 2562, 2563, 2564, 2565, 2567, 2569, 2570, 2571, 2572, 2573, 2576, 2579, 2581, 2583, 2586**

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Exam Duration: **90 Min.** **Maximum Marks: 50**

General instruction(s): **Answer ALL the questions**

Q. No.	Question	Max Marks
1.	**Obtain the PDNF and PCNF of the following formula and hence conclude whether it is a tautology ?** $(P \rightarrow (Q \wedge R)) \wedge (\neg P \rightarrow (\neg Q \wedge \neg R))$	**10**
	Let S= $(P \rightarrow (Q \wedge R)) \wedge (\neg P \rightarrow (\neg Q \wedge \neg R))$ $S = (\neg P \vee Q \vee \neg R) \wedge (\neg P \vee Q \vee R) \wedge (P \vee \neg Q \vee \neg R) \wedge (P \vee \neg Q \vee R)$ $\wedge (P \vee Q \vee \neg R) \wedge (\neg P \vee \neg Q \vee R)$ Which is PCNF of S To find PDNF $\neg(\neg S) = S = (P \wedge Q \wedge R) \vee (\neg P \wedge \neg Q \wedge \neg R)$ Which is PDNF of S Therefore the given statement formula is neither tautology nor contradiction and it is contingency statement..	
2.	**Verify whether the conclusion logically follows from the given arguments.** **"If you send me an e-mail message, then I will finish writing the program." "If you do not send me an e-mail message, then I will go to sleep early." "If I go to sleep early, then I will wake up feeling refreshed." Leads to the conclusion, "If I do not finish writing the program, then I will wake up feeling refreshed."**	**10**

Solution: Let p be the proposition "You send me an e-mail message," q the proposition "I will finish writing the program," r the proposition "I will go to sleep early," and s the proposition "I will wake up feeling refreshed." Then the premises are $p \rightarrow q$, $\neg p \rightarrow r$, and $r \rightarrow s$. The desired conclusion is $\neg q \rightarrow s$. We need to give a valid argument with premises $p \rightarrow q$, $\neg p \rightarrow r$, and $r \rightarrow s$ and conclusion $\neg q \rightarrow s$.

This argument form shows that the premises lead to the desired conclusion.

Step	Reason
1. $p \rightarrow q$	Premise
2. $\neg q \rightarrow \neg p$	Contrapositive of (1)
3. $\neg p \rightarrow r$	Premise
4. $\neg q \rightarrow r$	Hypothetical syllogism using (2) and (3)
5. $r \rightarrow s$	Premise
6. $\neg q \rightarrow s$	Hypothetical syllogism using (4) and (5)

3. **Show that the conclusion $(\forall x)\,(F(x) \rightarrow \neg S(x))$ follows logically from the premises $(\exists x)(F(x) \wedge S(x)) \rightarrow (\forall y)(M(y) \rightarrow W(y))$ and $(\exists y)(M(y) \wedge \neg W(y))$.** **10**

{1}	(1)	$(\exists y)[M(y) \wedge \neg W(y)]$	Rule P
{1}	(2)	$[M(z) \wedge \neg W(z)]$	Rule ES, (1)
{1}	(3)	$\neg[M(z) \rightarrow W(z)]$	Rule T, (2), and $\neg(P \rightarrow Q) \Leftrightarrow P \wedge \neg Q$
{1}	(4)	$(\exists y)\neg[M(y) \rightarrow W(y)]$	Rule EG, (3)
{1}	(5)	$\neg(y)[M(y) \rightarrow W(y)]$	Rule T, (4), and $\neg(x)A(x) \Leftrightarrow (\exists x)\neg A(x)$
{1}	(6)	$(\exists x)[F(x) \wedge S(x)] \rightarrow (y)[M(y) \rightarrow W(y)]$	Rule P
{1, 6}	(7)	$\neg(\exists x)[F(x) \wedge S(x)]$	Rule T, (5), (6) and I_{12}
{1, 6}	(8)	$(x)\neg[F(x) \wedge S(x)]$	Rule T, (7), and $\neg(x)A(x) \Leftrightarrow (\exists x)\neg A(x)$
{1, 6}	(9)	$\neg[F(z) \wedge S(z)]$	Rule US, (8)
{1, 6}	(10)	$\neg F(z) \vee \neg S(z)$	Rule T, (9), and De Morgan's laws
{1, 6}	(11)	$F(z) \rightarrow \neg S(z)$	Rule T, (10), and $P \rightarrow Q \Leftrightarrow \neg P \vee Q$
{1, 6}	(12)	$(x)(F(x) \rightarrow \neg S(x))$	Rule UG, (11)

Hence, the result.

4. **Define a subgroup with an example. Prove that "A non-empty subset H of a group G is a subgroup of (G, $*$) if and only if $a, b \in H \Rightarrow a * b^{-1} \in H$.** **10**

Sub Groups

Definition (7):

A non-empty subset H of a group G is said to be a subgroup of G if H itself is a group under the same operation defined on G and with the same identity element. (OR)

Let (G, $*$) be a group and H $\subseteq$ G (H $\neq \phi$) is said to be a subgroup (H, $*$) of (G, $*$) if

(i) $e \in$ H, where e is the identity of G

(ii) For any $a \in$ H, $a^{-1} \in$ H

(iii) For $a, b \in$ H, $a * b \in$ H.

Example (11) :

(i) The set of all integers is a subgroup of the set of all real numbers under addition. That is (Z, +) is a subgroup of (R, +)

(ii) The set of even integers $2Z = \{2k/k \in Z\}$ is a subgrop of (Z, +)

(iii) The set of non-negative integers is not a subgroup of (Z, +), because, except 0, no element have the additive inverse.

Proof : (Necessary Condition) :

Assume that H is a subgroup of G. Since H itself is a group ; we have $a, b \in H. \Rightarrow a * b \in H$ (closure). Also $b \in H \Rightarrow b^{-1} \in H$ (Inverse).

$\therefore a, b \in H \Rightarrow a, b^{-1} \in H \Rightarrow a * b^{-1} \in H$

Sufficient Condition : Let $a * b^{-1} \in H \rightarrow (1)$ for all $a, b \in H$ and $H \subseteq G$. we have to prove that H is a subgroup of G.

For,

(i) Identity : Let $a \in H, a \in H \Rightarrow a * a^{-1} \in H$

$\Rightarrow e \in H$ (by (1)).

Hence the identity e is the element of H.

(ii) Existene of inverse :

Let $e \in H, a \in H \Rightarrow e * a^{-1} \in H \Rightarrow a^{-1} \in H$ (by (1))

i.e., Every element of H has an inverse which is in H.

(iii) Closure :

Let $b \in H \Rightarrow b^{-1} \in H$

$\therefore$ For $a, b \in H \Rightarrow a, b^{-1} \in H \Rightarrow a * (b^{-1})^{-1} \in H$

$\Rightarrow a * b \in H$. $\therefore$ H is closed under the composition.

(iv) Associative : Since $H \subseteq G$, the elements of H are also the elements of G. Since the composition in G is associative, it must also be associative in H.

$\therefore$ H itself is a group for the composition in G.

i.e., H is a subgroup of G.

5.	**Let G denote the set of all matrices of the form $\begin{bmatrix} x & x \\ x & x \end{bmatrix}$ where $x \in R^+$. Prove that G is a group under matrix multiplication.**	**10**

To prove that G is a group under matrix multiplication, we need to verify that G satisfies the four group axioms:

1. **Closure**: If A and B are matrices in G, then their product AB must also be in G.
2. **Associativity**: Matrix multiplication is associative, so this property will hold for G as well.
3. **Identity**: There must be an identity element I in G such that for every matrix $A \in G$, $AI = IA = A$.
4. **Inverses**: For every matrix $A \in G$, there must be an inverse matrix $A^{-1} \in G$ such that $AA^{-1} = A^{-1}A = I$.

Let's verify each property for G:

1. Closure

Consider two matrices A and B in G. By definition, matrices in G are of the form

$$A = \begin{bmatrix} x & x \\ x & x \end{bmatrix} \text{ and } \quad B = \begin{bmatrix} y & y \\ y & y \end{bmatrix}$$

where $x, y \in \mathbb{R}^+$.

To show closure, we need to compute the product AB and check if it is of the same form as matrices in G.

$$AB = \begin{bmatrix} x & x \\ x & x \end{bmatrix} \begin{bmatrix} y & y \\ y & y \end{bmatrix} = \begin{bmatrix} x \cdot y + x \cdot y & x \cdot y + x \cdot y \\ x \cdot y + x \cdot y & x \cdot y + x \cdot y \end{bmatrix} = \begin{bmatrix} 2xy & 2xy \\ 2xy & 2xy \end{bmatrix}$$

This resulting matrix is of the form $\begin{bmatrix} z & z \\ z & z \end{bmatrix}$ where $z = 2xy \in \mathbb{R}^+$. Therefore, AB is also in G, proving that G is closed under matrix multiplication.

2. Associativity

Matrix multiplication is associative in general, so this property holds for G as well.

3. Identity

We need to find an identity matrix $I \in G$ such that $AI = IA = A$ for any $A \in G$. Consider

$$I = \begin{bmatrix} e & e \\ e & e \end{bmatrix}$$

where e is a positive real number. To satisfy $AI = IA = A$ for any $A \in G$, we choose $e = 1$. Thus,

$$I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Let's verify:

$$AI = \begin{bmatrix} x & x \\ x & x \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} x \cdot 1 + x \cdot 1 & x \cdot 1 + x \cdot 1 \\ x \cdot 1 + x \cdot 1 & x \cdot 1 + x \cdot 1 \end{bmatrix} = \begin{bmatrix} 2x & 2x \\ 2x & 2x \end{bmatrix}$$

This doesn't match A, so I is not the correct choice. We need I to be in the form $\begin{bmatrix} z & z \\ z & z \end{bmatrix}$, where the entries of I should be such that $e = \frac{1}{2}$. Therefore, the correct identity matrix should be:

$$I = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Verify:

$$AI = \begin{bmatrix} x & x \\ x & x \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} x \cdot \frac{1}{2} + x \cdot \frac{1}{2} & x \cdot \frac{1}{2} + x \cdot \frac{1}{2} \\ x \cdot \frac{1}{2} + x \cdot \frac{1}{2} & x \cdot \frac{1}{2} + x \cdot \frac{1}{2} \end{bmatrix} = \begin{bmatrix} x & x \\ x & x \end{bmatrix}$$

Thus, I acts as the identity matrix in G.

4. Inverses

For a matrix $A = \begin{bmatrix} x & x \\ x & x \end{bmatrix}$ in G, we need to find its inverse A^{-1} such that $AA^{-1} = A^{-1}A = I$.

Let's assume A^{-1} has the form $\begin{bmatrix} a & a \\ a & a \end{bmatrix}$. We need

$$\begin{bmatrix} x & x \\ x & x \end{bmatrix} \begin{bmatrix} a & a \\ a & a \end{bmatrix} = \begin{bmatrix} x \cdot a + x \cdot a & x \cdot a + x \cdot a \\ x \cdot a + x \cdot a & x \cdot a + x \cdot a \end{bmatrix} = \begin{bmatrix} 2xa & 2xa \\ 2xa & 2xa \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

To satisfy this, we need:

$$2xa = \frac{1}{2} \quad \text{so} \quad a = \frac{1}{4x}$$

Thus, the inverse matrix is:

$$A^{-1} = \begin{bmatrix} \frac{1}{4x} & \frac{1}{4x} \\ \frac{1}{4x} & \frac{1}{4x} \end{bmatrix}$$

We can verify:

$$AA^{-1} = \begin{bmatrix} x & x \\ x & x \end{bmatrix} \begin{bmatrix} \frac{1}{4x} & \frac{1}{4x} \\ \frac{1}{4x} & \frac{1}{4x} \end{bmatrix} = \begin{bmatrix} \frac{x}{4x} + \frac{x}{4x} & \frac{x}{4x} + \frac{x}{4x} \\ \frac{x}{4x} + \frac{x}{4x} & \frac{x}{4x} + \frac{x}{4x} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Thus, the inverse matrix exists and is of the same form as matrices in G.

Thus, the inverse matrix exists and is of the same form as matrices in G.

Conclusion

Since G satisfies closure, associativity, the existence of an identity element, and the existence of inverses, G is indeed a group under matrix multiplication.